PRACTICAL NO. 1

Aim:- Getting started with Tableau and building basic visualization.

Objectives:-

The Objective of practical is to learn:

- How to connect data to Tableau.
- How Tableau organizes data.
- How to create different types of charts.
- How to build a dashboard using multiple charts.

Before starting, remember the workflow:

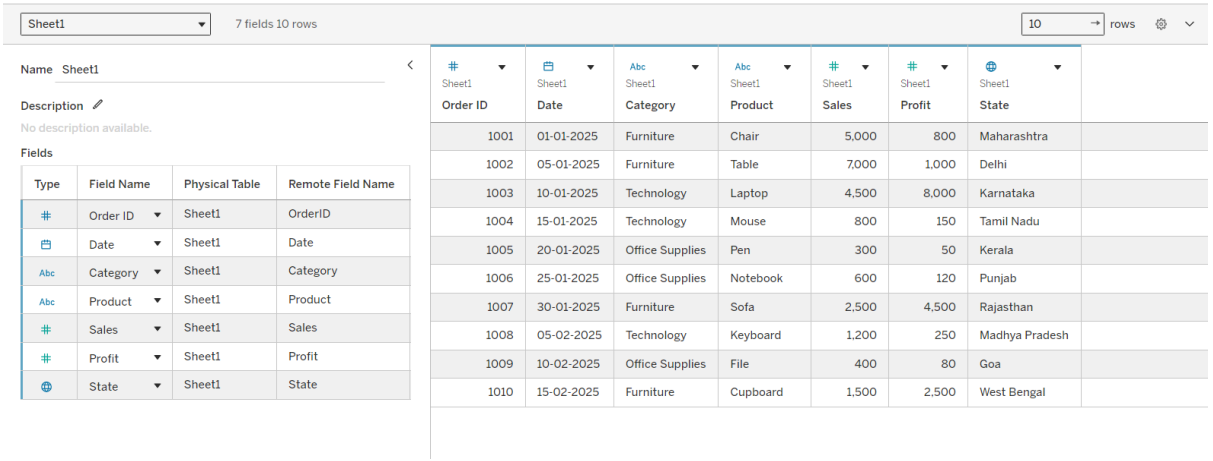
Excel/CSV Data
↓
Connect to Tableau
↓
Analyze Data
↓
Create Charts
↓
Build Dashboard

PRACTICAL NO. 1(A)

Aim:- Connect to a CSV Dataset and explore the Tableau interface.

Steps to Load a Dataset in Tableau Public:-

1. Open Tableau Public on your computer.
2. On the start page, click Connect to Data.
3. Select the appropriate data source (e.g., Microsoft Excel, Text File (CSV), JSON, or another supported format).
4. Browse to the location where your dataset is stored and select the file.
5. Click Open to import the dataset into Tableau Public.
6. The Data Source page will appear, where you can preview the data.



Sheet1 | 7 fields 10 rows | 10 rows

Name: Sheet1

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field Name
#	Order ID	Sheet1	OrderID
📅	Date	Sheet1	Date
Abc	Category	Sheet1	Category
Abc	Product	Sheet1	Product
#	Sales	Sheet1	Sales
#	Profit	Sheet1	Profit
🌐	State	Sheet1	State

Order ID	Date	Category	Product	Sales	Profit	State
1001	01-01-2025	Furniture	Chair	5,000	800	Maharashtra
1002	05-01-2025	Furniture	Table	7,000	1,000	Delhi
1003	10-01-2025	Technology	Laptop	4,500	8,000	Karnataka
1004	15-01-2025	Technology	Mouse	800	150	Tamil Nadu
1005	20-01-2025	Office Supplies	Pen	300	50	Kerala
1006	25-01-2025	Office Supplies	Notebook	600	120	Punjab
1007	30-01-2025	Furniture	Sofa	2,500	4,500	Rajasthan
1008	05-02-2025	Technology	Keyboard	1,200	250	Madhya Pradesh
1009	10-02-2025	Office Supplies	File	400	80	Goa
1010	15-02-2025	Furniture	Cupboard	1,500	2,500	West Bengal

PRACTICAL NO. 1(B)

Aim:- Identify and differentiate Measures and Dimensions.

Steps to Check Measures and Dimensions in Tableau Public

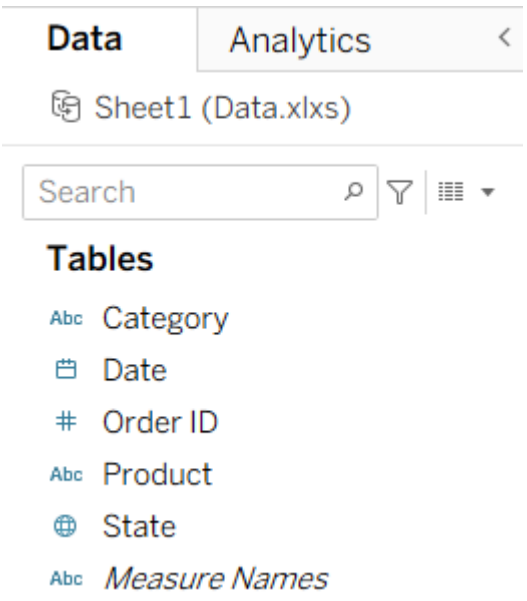
1. Open Tableau Public and load your dataset.
2. Navigate to the Worksheet (Sheet 1).
3. Look at the Data Pane on the left side of the screen.
4. The fields are automatically divided into two sections:

Usually Dimension are in blue color.

Example:

1) Category
2) Date
3) Order ID
4) Product
5) State

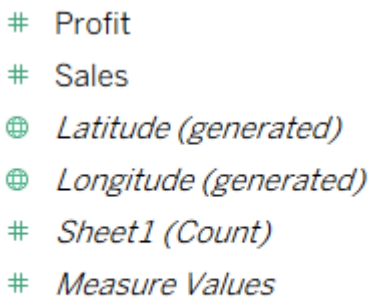
These describes the data.



Measures are generally green in color.

Example:

1) Profit
2) Sales



These measures can be calculated.

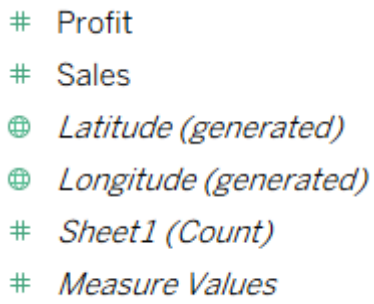
Example:

1) SUM(Sales)
2) AVG(Profit)
3) COUNT(Order ID)

PRACTICAL NO. 1(C)

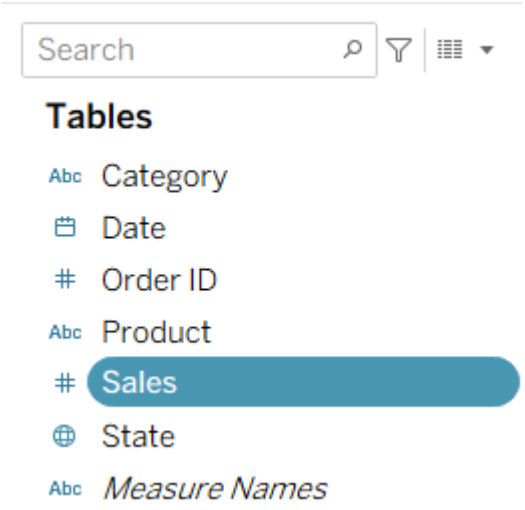
Aim:- Convert fields between discrete and continuous and observe changes.

Suppose “Sales” is under Dimensions



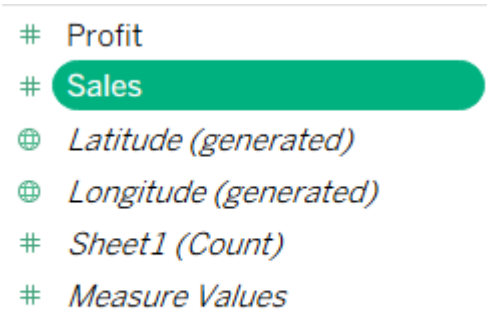
Convert Measures to Dimension:

1) Right - Click on Sales.
2) Click
3) Click on Convert to Dimension.
4) Now it moves under Dimension.



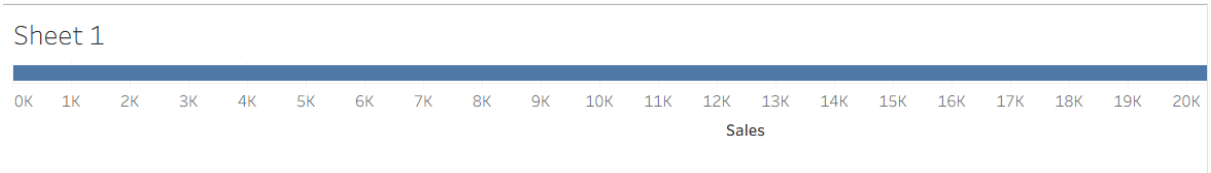
Now, Convert Dimension back to Measures:

1) Right Click on Sales.
2) Click.
3) Click on convert to Measures.
4) Observe.
5) It moves again to Measures.



Continuous & Discrete:

1) Drag Sales into Column.
2) Click the small arrow on Sales.
3) Click on Continuous
4) Observe
5) Green Pill
6) Axis becomes Continue



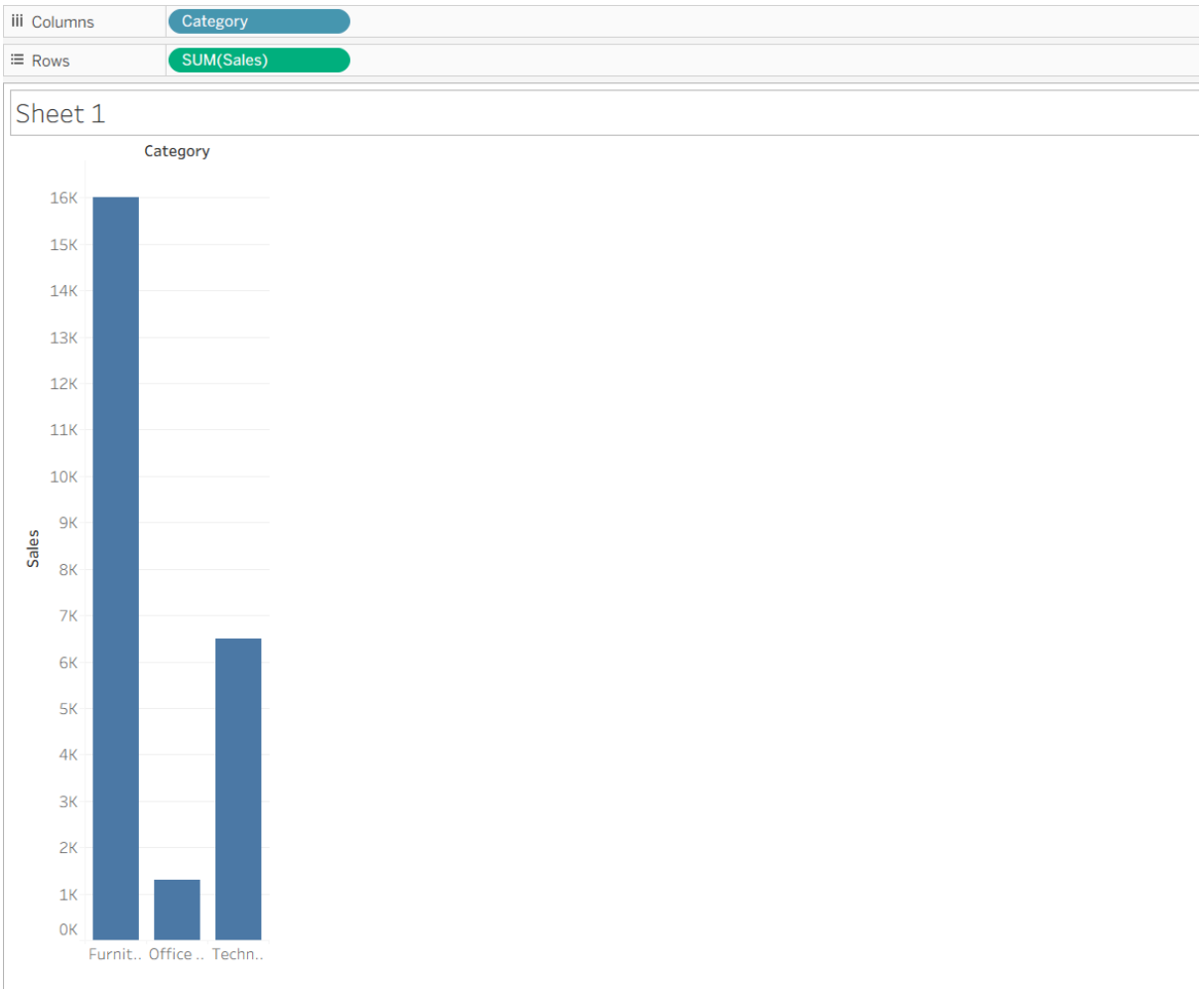
7) Now click on Discrete
8) Observe
9) Blue Pill
10) Separate header appear



PRACTICAL NO. 1(D)

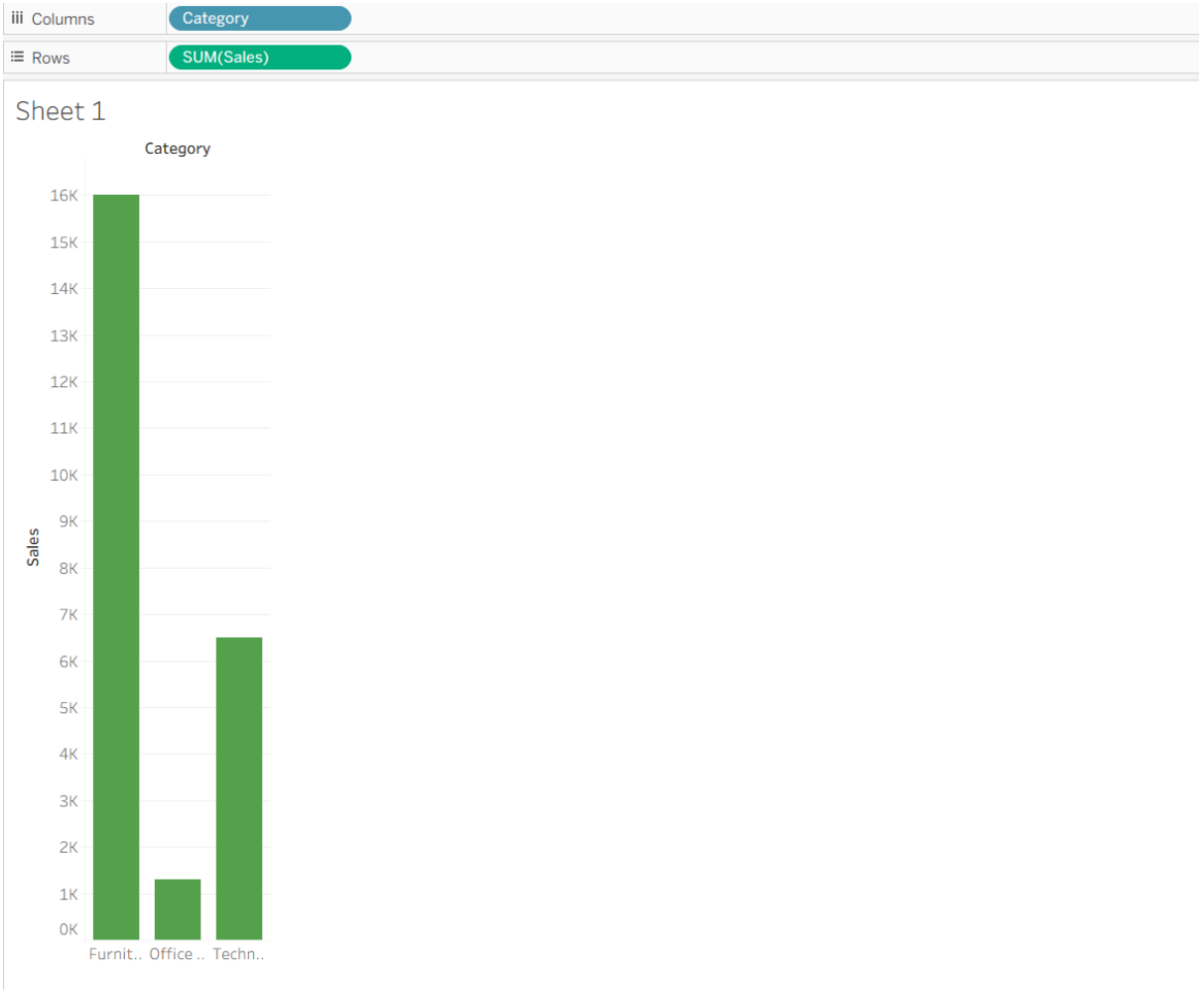
Aim:- Create and customize a basic bar chart.

Drag “Category” to Column and Drag “Sales” to row immediately bars appear.



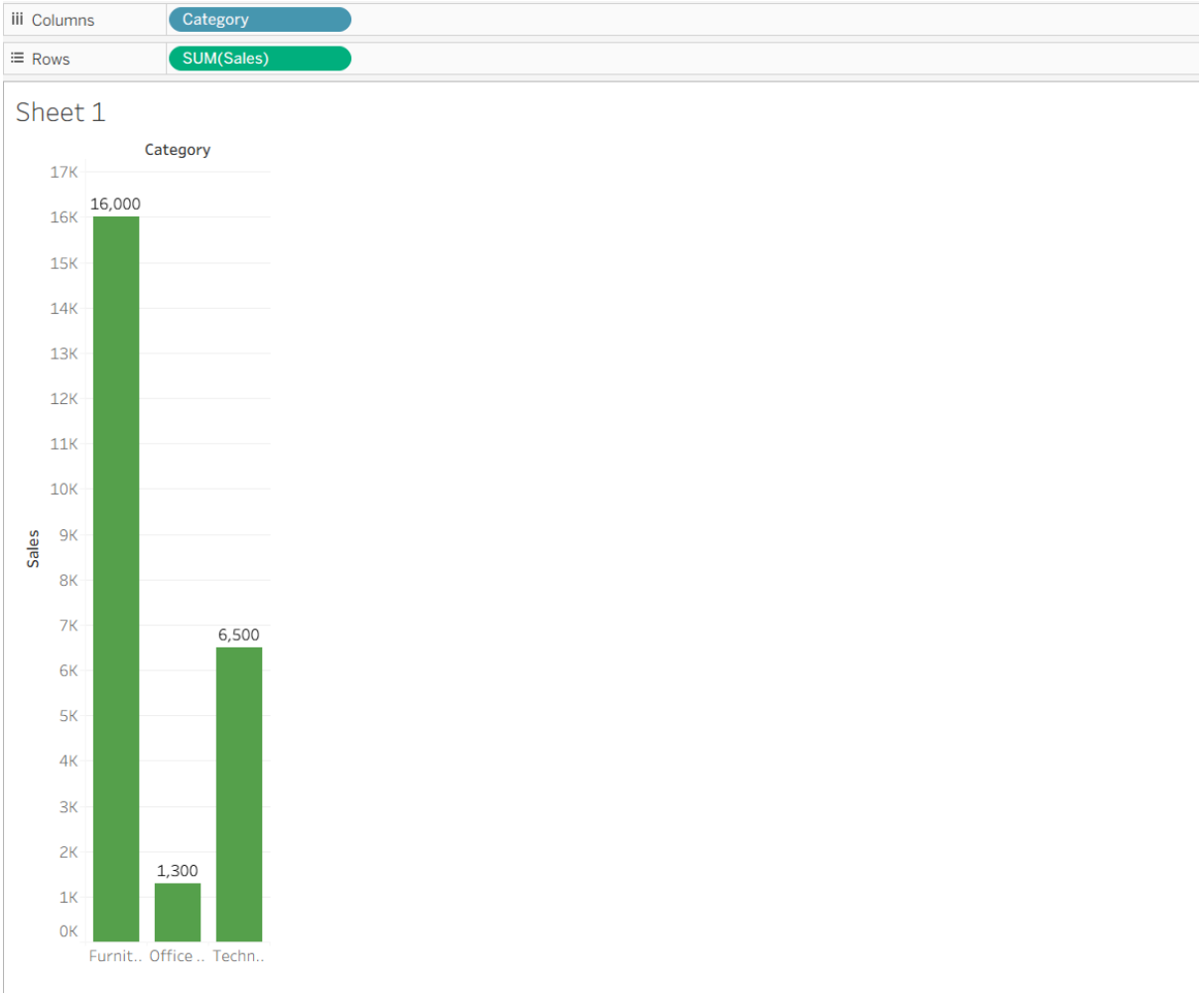
Customize:-

1) Click
2) Color
3) Choose any Color



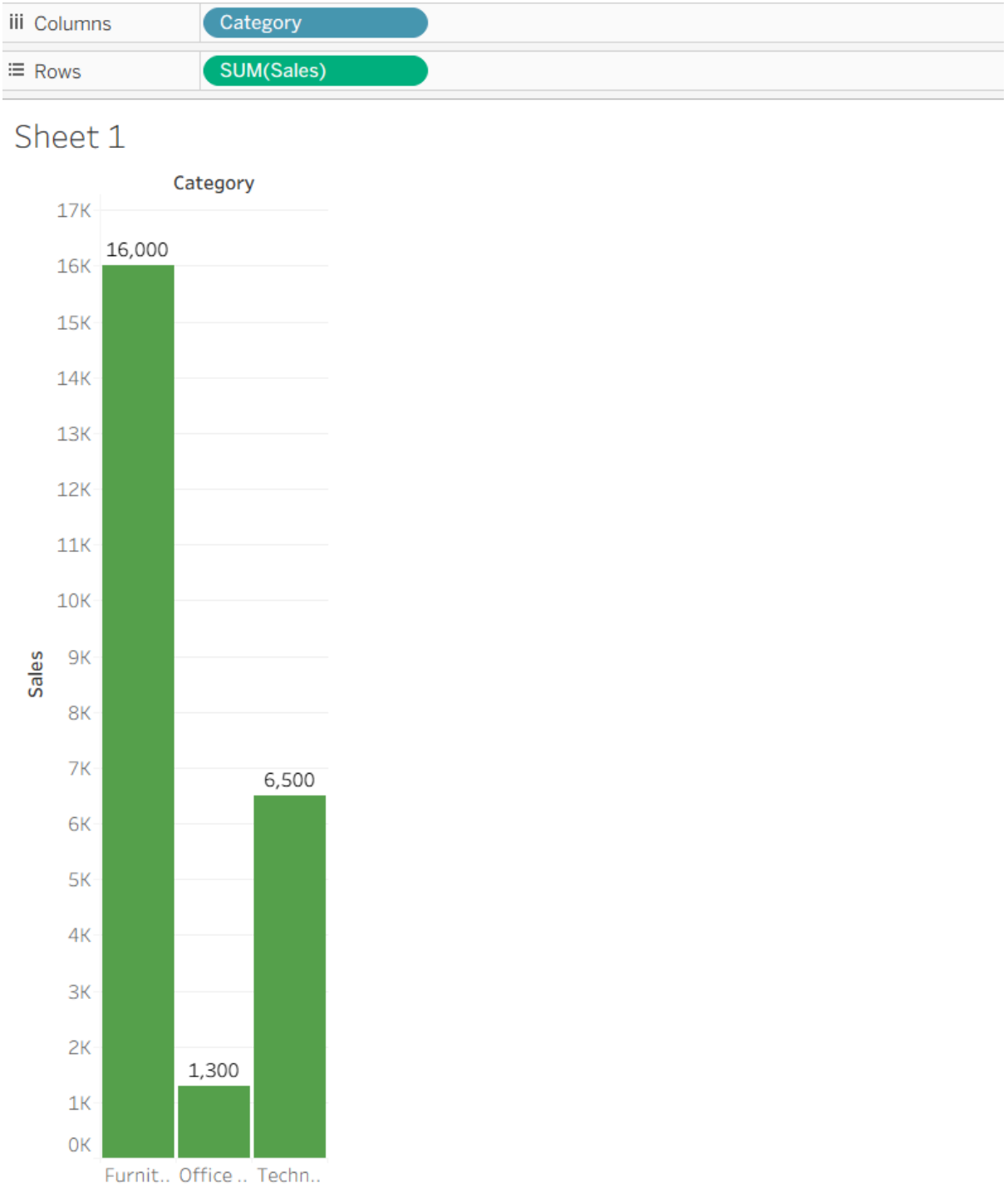
Add MarkUp Labels:-

1) Click
2) Label
3) Check
4) Show Mark Labels
5) Sales value appears



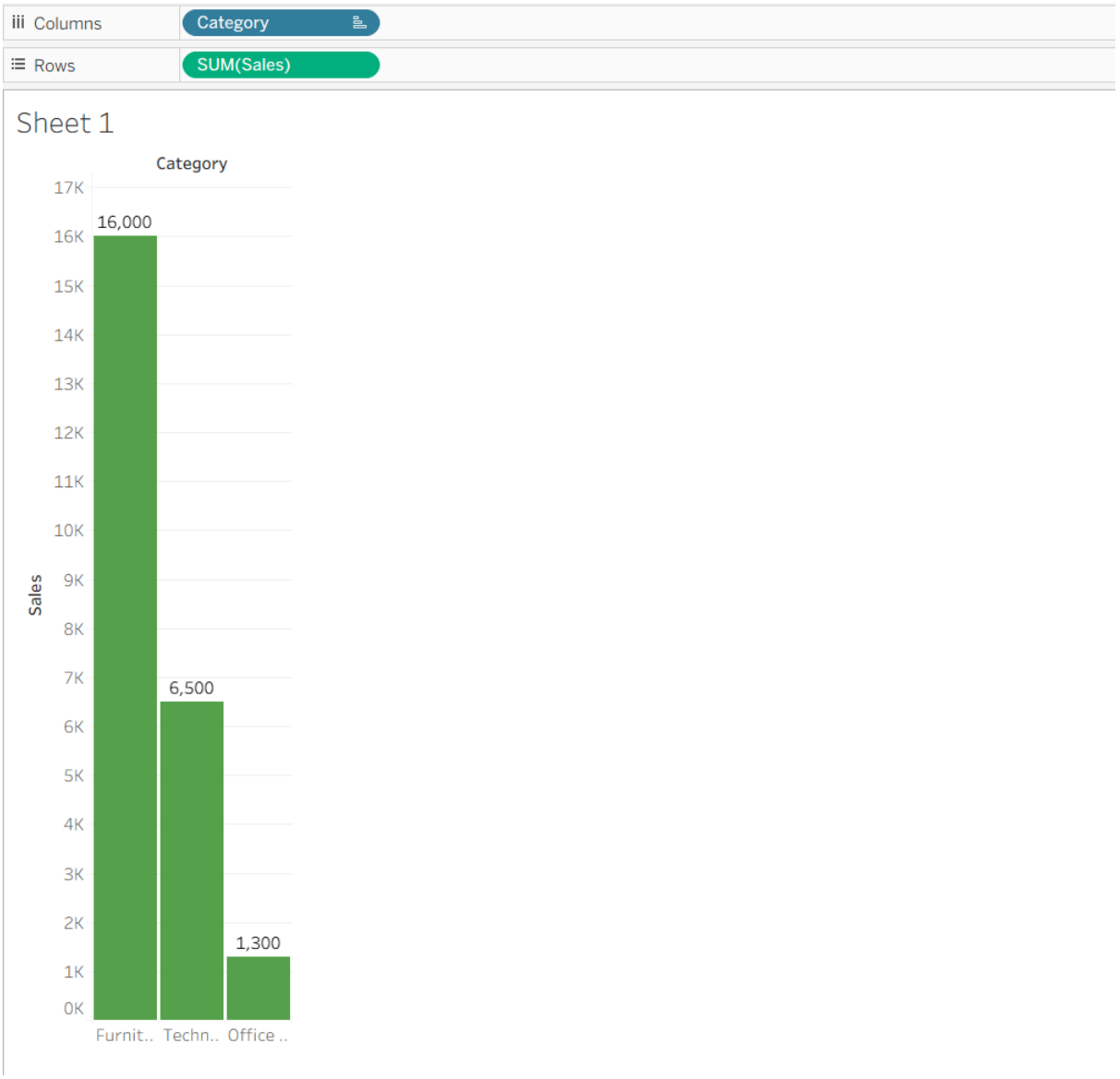
Increase Size:-

1) Click
2) Size
3) Increase Size



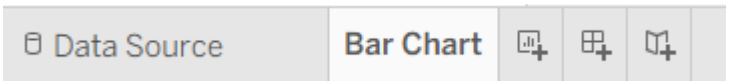
Sorting:-

1) Click
2) Sort Descending
3) Bars arrange from highest to lowest



Rename Sheet:-

1) Double Click on “Sheet1”.
2) Click on rename
3) Enter Name “Bar Chart”.



PRACTICAL NO. 1(E)

Aim:- Build a line chart for time-series analysis.

1) Open new Worksheet.
2) Drag Date to Column.
3) If Tableau asks Year, Month, Day choose Month.
4) Now drag Sales to choose mark line drag Sales to Rows shelf.
5) Now monthly trend appears.



PRACTICAL NO. 1(F)

Aim:- Create a geographic map visualization.

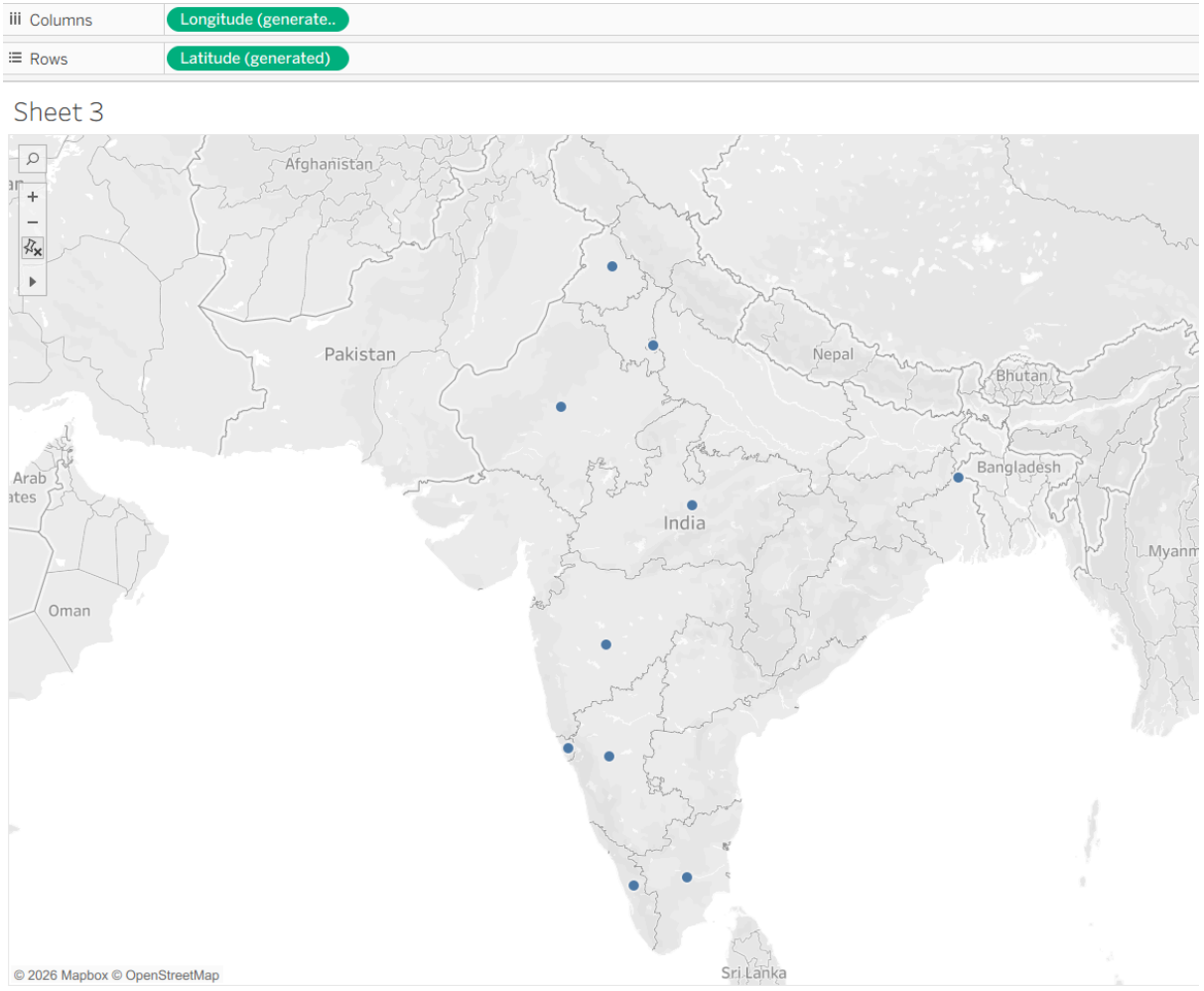
1) Open new worksheet.
2) Drag State to worksheet.
3) If Tableau recognizes state, it creates latitude and longitude automatically.

If not:

1) Right click on State
2) Click on geographic role
3) Then click on State/Province
4) Then double state.

Step 1: Create the Map

1) Drag State to worksheet
2) Tableau recognizes it as geographic field and display a map.
   (It it dosen't right click on State → Geographic Role → State/Province)

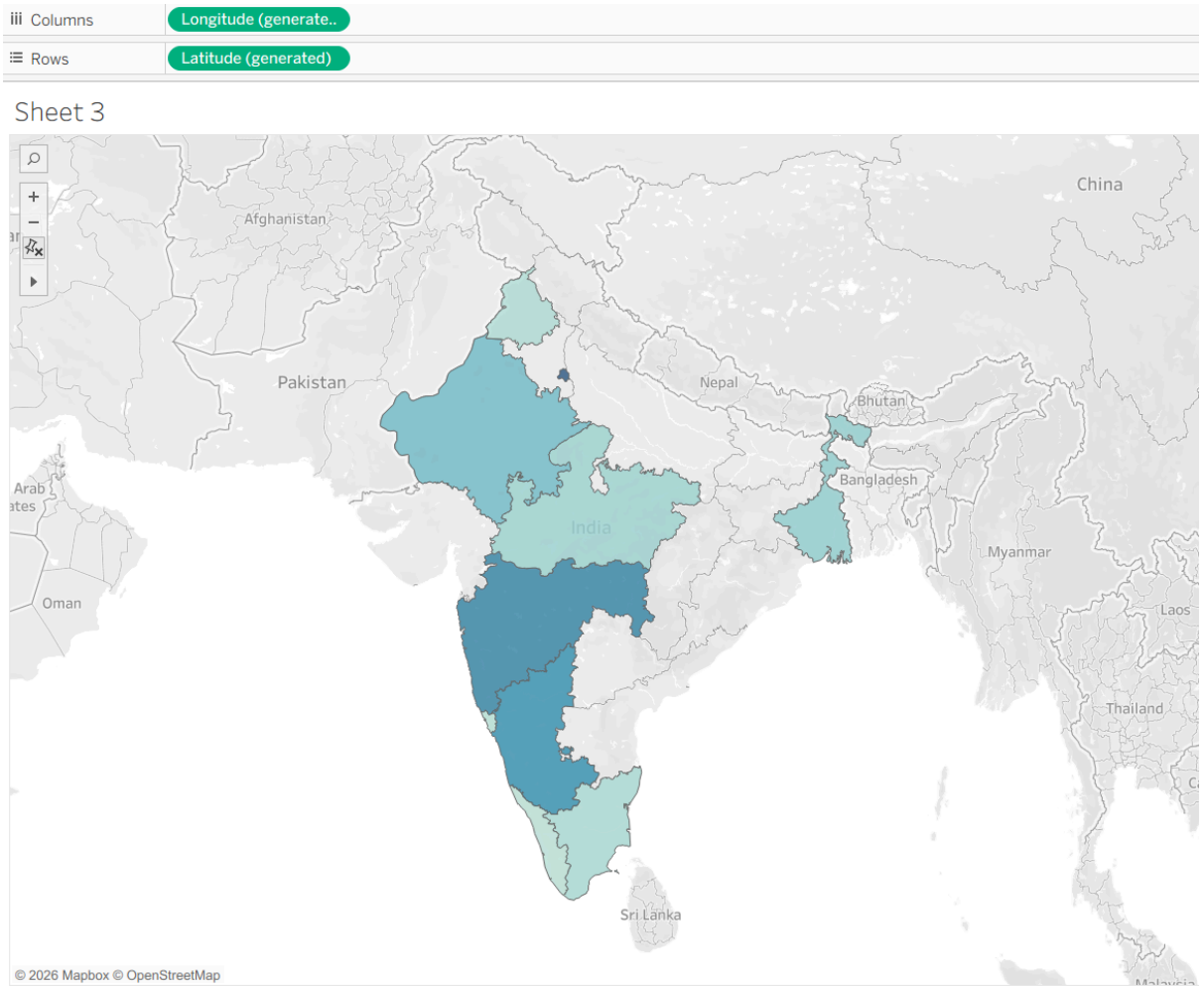


Step 2: Drag Sales to Color

1) In the data pane(left side), find Sales.
2) Click and hold Sales.
3) Drag it to color option on the mark card.
4) Release the mouse.

The Map color will changes automatically:

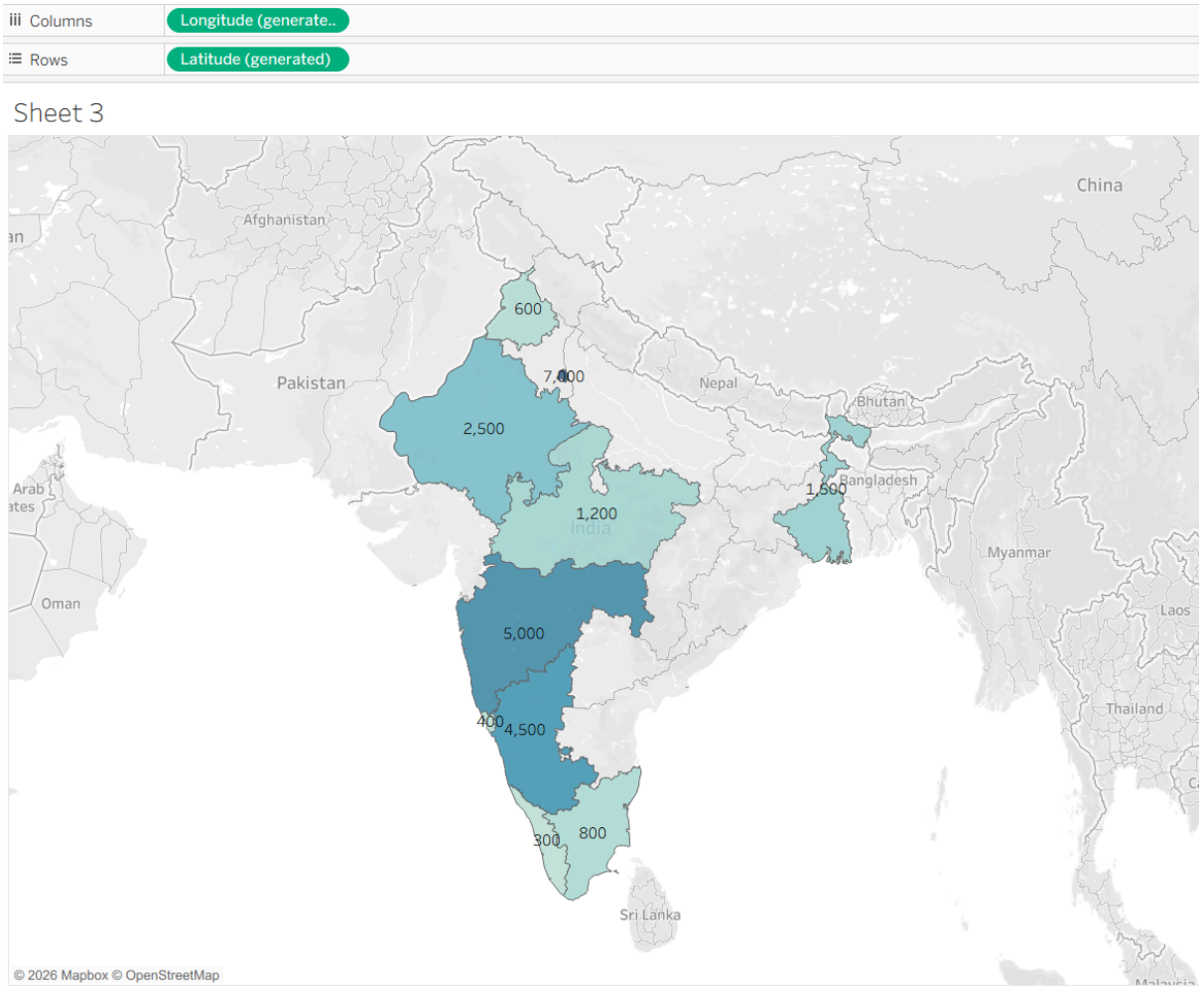
1) Darker color = Higher Sales
2) Lighter color = Lower Sales



Step 3: Drag Sales to Label

1) Again, click and hold Sales from the data pane.
2) Drag it to label on the mark card.
3) Release the mouse.

Each state on the map will display its value.



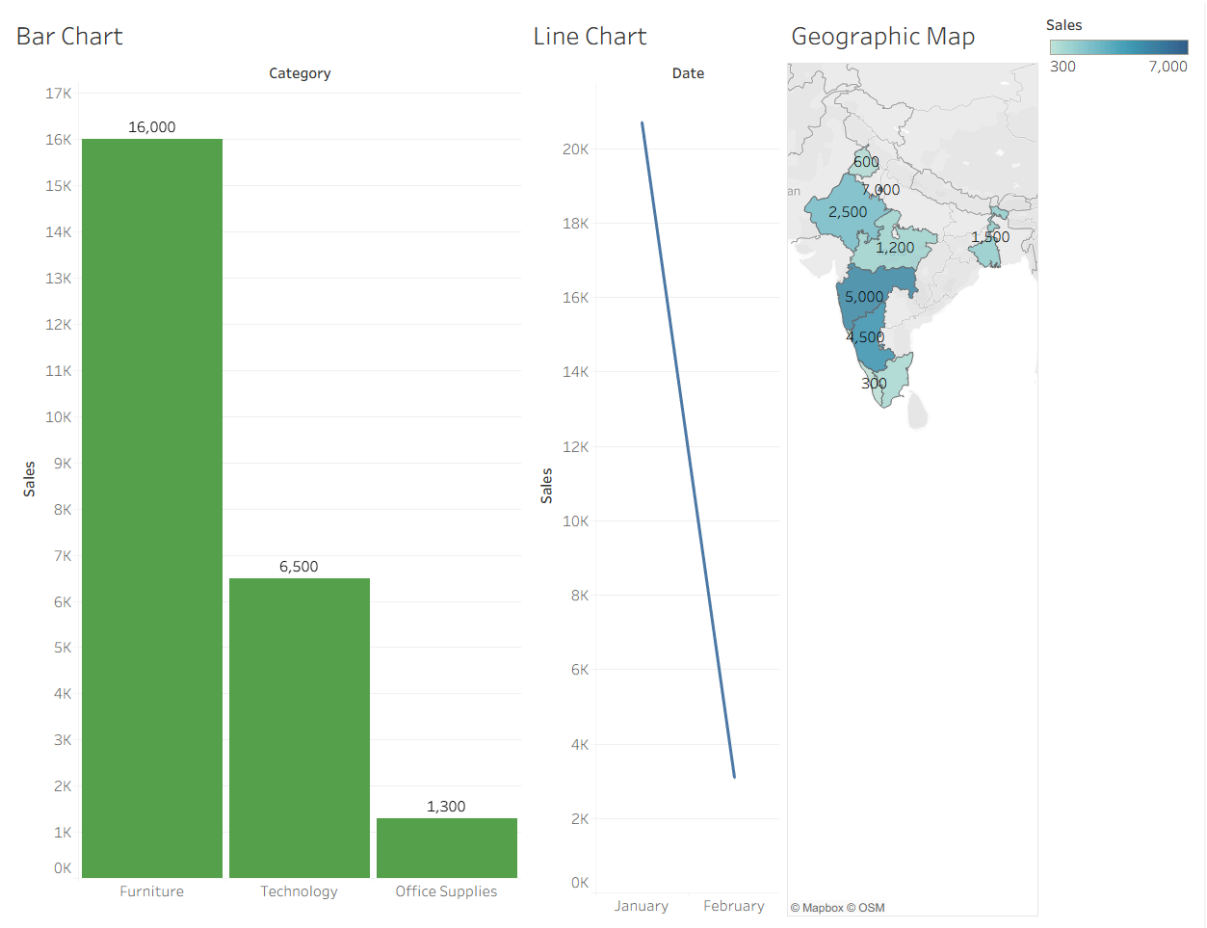
PRACTICAL NO. 1(F)

Aim:- Combine multiple charts into simple interactive dashboard.

Create Dashboard:

1) Click on New Dashboard
2) (Bottom tab with dashboard icon)
3) Empty dashboard opens
4) Left side contains all sheets
5) Drag all those sheets in dashboard
    a) Bar Chart
    b) Line Chart
    c) Geographic Map

Arrange them neatly.



Conclusion:

In this practical, we learned the fundamentals of using Tableau for data visualization and analysis. We successfully connected a CSV dataset, explored the Tableau interface, and understood the difference between Measures and Dimensions. We also observed how converting fields between discrete and continuous affects the visualization. Various charts, including bar charts, line charts, and geographic maps, were created and customized to represent data effectively. Finally, multiple visualizations were combined into an interactive dashboard, enabling users to analyze data from different perspectives in a single view. This practical provided a strong foundation in Tableau and demonstrated how visual analytics can transform raw data into meaningful insights.